Capturing non-Markovian effects in weather prediction models

1 Introduction

Many numerical and learned forecasters advance a resolved weather state from its current value or a short window of recent values. Examples include the numerical Integrated Forecasting System (IFS) configuration used to produce ERA5 ECMWF [2016], Hersbach et al. [2020], and learned Pangu-Weather Bi et al. [2023], GraphCast Lam et al. [2023], FuXi Chen et al. [2023], ECMWF's ERA5-native-grid AIFS Lang et al. [2024], WeatherNext 2 Alet et al. [2025], and WeatherNext 3 Rasp et al. [2026]. This is a practical representation for autoregressive prediction, but the resolved variables need not contain every influence on their subsequent evolution. In global weather prediction, finite horizontal resolution, limited observed variables, and approximate representations of smaller-scale processes all constrain the state available to a forecast model. Whether information about omitted processes can be recovered from the history of that resolved state is therefore a question about the chosen model representation, rather than a claim that the full atmospheric dynamics lack a Markovian state.

The difficulty of resolving all relevant scales is familiar from turbulent-flow simulation. Direct numerical simulation (DNS) discretizes the governing equations finely enough to represent dynamically active scales, and its grid requirements grow rapidly with Reynolds number. For example, a grid-point estimate for a spatially developing turbulent boundary layer gives $N_{\mathrm{grid}} \sim Re_{L_x}^{2.05}$ under locally adapted DNS spacing Yang and Griffin [2021]. This is a cost estimate for that flow, not a forecast-error law for the atmosphere. Resolution also affects what a simulation can represent: insufficient small-scale resolution can leave mean dissipation and enstrophy nearly converged while strongly underestimating higher-order gradient statistics and distribution tails Donzis et al. [2008]. Coarser practical models must represent the effects of unresolved motions through an effective closure.

Such a closure need not depend only on the present resolved state. Projecting full dynamics onto a smaller set of variables can yield an instantaneous contribution, a contribution depending on the history of those variables, and a term associated with unobserved initial conditions. The Mori–Zwanzig formalism makes this decomposition explicit; data-driven studies have learned its memory contribution in reduced dynamical systems Lin et al. [2021], De Wit et al. [2026]. This motivates temporal memory as a possible way to represent effects left out by a finite-window predictor. It does not imply that a learned latent state can be identified with a unique physical memory kernel.

For the Kuramoto–Sivashinsky equation, the advantage of memory over a memoryless operator changes with grid resolution, and the crossover depends on viscosity and the associated high-frequency content Buitrago Ruiz et al. [2025]. Whether an analogous crossover exists for global weather forecasting, and at what grid resolution it might occur, remains an empirical question.

Here, we take a first step toward addressing this question by testing whether temporal memory can improve GraphCast forecasts at 1° resolution. We augment a frozen GraphCast model Lam et al. [2023] with a trainable Mamba correction branch that combines spatial message passing with Mamba layers Gu and Dao [2023]. This branch carries a temporal memory state across autoregressive forecast steps and adds a correction to each GraphCast prediction.

Preprint.

Learning a residual correction focuses training on what the pretrained predictor misses. Freezing the predictor leaves only the correction branch's parameters to optimize. The Mori–Zwanzig decomposition offers a structural motivation for this *additive* ansatz: a local-in-time contribution and a history-dependent contribution enter the resolved evolution as separate terms Lin et al. [2021], De Wit et al. [2026]. The outputs are added linearly, while the learned memory map itself can be nonlinear.

Output corrections to frozen time-series forecasters have been studied in δ-Adapter Liang et al. [2026], and neural corrections have been applied during the integration of a numerical weather model Chen et al. [2022]. Memory within a forecaster is also established, including recurrent global weather models Hu et al. [2023] and the neural-operator approach above. Our focus is on temporal memory in a separate correction branch added to a frozen pretrained forecaster.

We study the general output-level form

$$(c_t, m_t) = R_\phi(X_t, m_{t-1}), \qquad \widehat{Y}_t = G_\theta(X_t) + c_t, \qquad \theta \text{ fixed}, \tag{1}$$

where G_θ is the pretrained GraphCast backbone, R_ϕ is the trainable Mamba correction branch, c_t is its forecast correction, and m_t is its temporal memory state. This state persists between forecast calls and comprises all state-space model (SSM) states and causal-convolution caches in the branch. In our implementation, G_θ is GraphCast and R_ϕ is the correction branch described above. The predictor and correction meet at the output, so this formulation does not require access to the predictor's internal layers. A zero-initialized output head makes the GraphCast–Mamba model start from the GraphCast forecast; only the correction branch is then trained.

We test this instantiation at 1° resolution on 32 matched forecast initializations from a development set. The selected correction reduces the GraphCast normalized weighted mean-squared error (GraphCast loss) by 16.34% over ten-day rollouts and 23.68% at day 10 (240 h after initialization). Separate memory-reset controls, measured with an approximate reconstruction of that loss, support a contribution from persistent temporal context beyond the branch's additional trainable capacity. These results establish a case for memory-based correction of a strong pretrained forecaster at the tested resolution. The resolution at which its memory benefit might diminish, and whether the same correction strategy transfers to other forecasters, require separate matched experiments.

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